Supplementary Methods

Table 1: NHANES Variables and Descriptions

Variable	Description
SurveyYr	Survey cycle year.
ID	Participant identifier.
Gender	Sex of participant (male/female).
Age	Age in years; ages 80 coded as 80.
AgeDecade	Age grouped in 10-year intervals.
AgeMonths	Age in months for younger participants.
Race1	Race/ethnicity: Mexican, Hispanic, White, Black, Other.
Race3	Race including Asian category (not available in 2009–10).
Education	Highest educational attainment.
MaritalStatus	Current marital status.
HHIncome	Annual household income categories.
HHIncomeMid	Midpoint of income category (numeric).
Poverty	Family income-to-poverty ratio.
HomeRooms	Number of rooms in the home.
HomeOwn	Housing tenure: own, rent, or other.
Work	Employment status.
Weight	Body weight (kg).
Length	Recumbent length (cm) for young children.
HeadCirc	Head circumference (cm).
Height	Standing height (cm).
BMI	Body mass index (kg/m^2).
BMICatUnder20yrs	BMI category for ages 2–19.
BMI_WHO	BMI category based on WHO cutoffs.
Pulse	Pulse rate (beats per minute).
BPSysAve	Average systolic blood pressure.
BPDiaAve	Average diastolic blood pressure.
BPSys1	First systolic blood pressure reading.
BPSys2	Second systolic blood pressure reading.

Variable	Description
BPSys3	Third systolic blood pressure reading.
BPDia1	First diastolic blood pressure reading.
BPDia2	Second diastolic blood pressure reading.
BPDia3	Third diastolic blood pressure reading.
Testosterone	Total testosterone level.
DirectChol	Direct HDL cholesterol (mmol/L).
TotChol	Total cholesterol (mmol/L).
UrineVol1/2	Urine volume from first/second test.
UrineFlow1/2	Urine flow rate (mL/min).
Diabetes	Self-reported doctor-diagnosed diabetes (Yes/No).
DiabetesAge	Age at first diabetes diagnosis.
HealthGen	Self-rated general health.
DaysPhysHlthBad	Days physical health was 'not good' in past 30 days.
DaysMentHlthBad	Days mental health was 'not good' in past 30 days.
LittleInterest	Lack of interest in activities (PHQ item).
Depressed	Feeling depressed or hopeless (PHQ item).
nPregnancies	Number of pregnancies (females only).
nBabies	Number of live births.
PregnantNow	Pregnancy status at exam.
Age1stBaby	Age at first live birth.
SleepHrsNight	Hours of sleep per night.
SleepTrouble	Ever told by a provider of sleep trouble (Yes/No).
PhysActive	Participation in moderate/vigorous activity (Yes/No).
PhysActiveDays	Days per week physically active.
TVHrsDay	Hours of TV watching per day.
CompHrsDay	Hours of computer/device use per day.
TVHrsDayChild	TV hours per day (children).
CompHrsDayChild	Computer use per day (children).
Alcohol12PlusYr	Consumed 12 alcoholic drinks in the past year (Yes/No).
AlcoholDay	Drinks consumed on drinking days.
AlcoholYear	Days drank alcohol in past year.
SmokeNow	Currently smokes (if smoked 100 lifetime cigarettes).
Smoke100	Smoked at least 100 cigarettes in lifetime (Yes/No).
SmokeAge	Age started smoking regularly.
Marijuana	Ever used marijuana.
AgeFirstMarij	Age first used marijuana.
RegularMarij	Regular marijuana use (monthly for 1 year).
AgeRegMarij	Age started regular marijuana use.
HardDrugs	Ever used cocaine, heroin, meth, etc.
SexEver	Ever had vaginal, oral, or anal sex.
SexAge	Age at first sexual intercourse.

Variable	Description
SexNumPartnLife	Number of lifetime sexual partners.
SexNumPartYear	Number of partners in past 12 months.
SameSex	Ever had same-sex partner (Yes/No).
SexOrientation	Self-reported sexual orientation.

Table 2: Missing Data Summary for All Variables

Variable	Missing (n)	Total N	Missing (%)
AgeRegMarij	2495	2873	86.8
UrineVol2	2488	2873	86.6
UrineFlow2	2489	2873	86.6
DiabetesAge	2453	2873	85.4
AgeFirstMarij	2086	2873	72.6
Age1stBaby	1910	2873	66.5
nBabies	1731	2873	60.3
nPregnancies	1675	2873	58.3
Testosterone	1651	2873	57.5
PhysActiveDays	1572	2873	54.7
SmokeAge	1563	2873	54.4
Race3	1539	2873	53.6
TVHrsDay	1541	2873	53.6
CompHrsDay	1539	2873	53.6
SmokeNow	1509	2873	52.5
SexOrientation	1507	2873	52.5
SexNumPartYear	1472	2873	51.2
Marijuana	1466	2873	51.0
RegularMarij	1466	2873	51.0
AgeMonths	1456	2873	50.7
AlcoholDay	1094	2873	38.1
SexAge	987	2873	34.4
SexNumPartnLife	955	2873	33.2
HardDrugs	934	2873	32.5
SexEver	933	2873	32.5
SameSex	933	2873	32.5
AlcoholYear	576	2873	20.0
LittleInterest	290	2873	10.1
Depressed	290	2873	10.1
Alcohol12PlusYr	276	2873	9.6
DaysPhysHlthBad	271	2873	9.4
DaysMentHlthBad	268	2873	9.3

Variable	Missing (n)	Total N	Missing (%)
HealthGen	264	2873	9.2
HHIncome	261	2873	9.1
HHIncomeMid	261	2873	9.1
AgeDecade	249	2873	8.7
Poverty	232	2873	8.1
BPSys1	221	2873	7.7
BPDia1	221	2873	7.7
UrineFlow1	199	2873	6.9
BPSys2	177	2873	6.2
BPDia2	177	2873	6.2
BPSys3	174	2873	6.1
BPDia3	174	2873	6.1
DirectChol	164	2873	5.7
TotChol	164	2873	5.7
BPSysAve	104	2873	3.6
BPDiaAve	104	2873	3.6
Pulse	97	2873	3.4
BMI_WHO	54	2873	1.9
UrineVol1	50	2873	1.7
BMI	35	2873	1.2
Height	32	2873	1.1
Weight	29	2873	1.0
HomeRooms	17	2873	0.6
HomeOwn	14	2873	0.5
SleepHrsNight	8	2873	0.3
Education	7	2873	0.2
MaritalStatus	4	2873	0.1
Gender	0	2873	0.0
Age	0	2873	0.0
Race1	0	2873	0.0
Work	0	2873	0.0
Diabetes	1	2873	0.0
SleepTrouble	0	2873	0.0
PhysActive	0	2873	0.0
Smoke100	0	2873	0.0
Smoke100n	0	2873	0.0

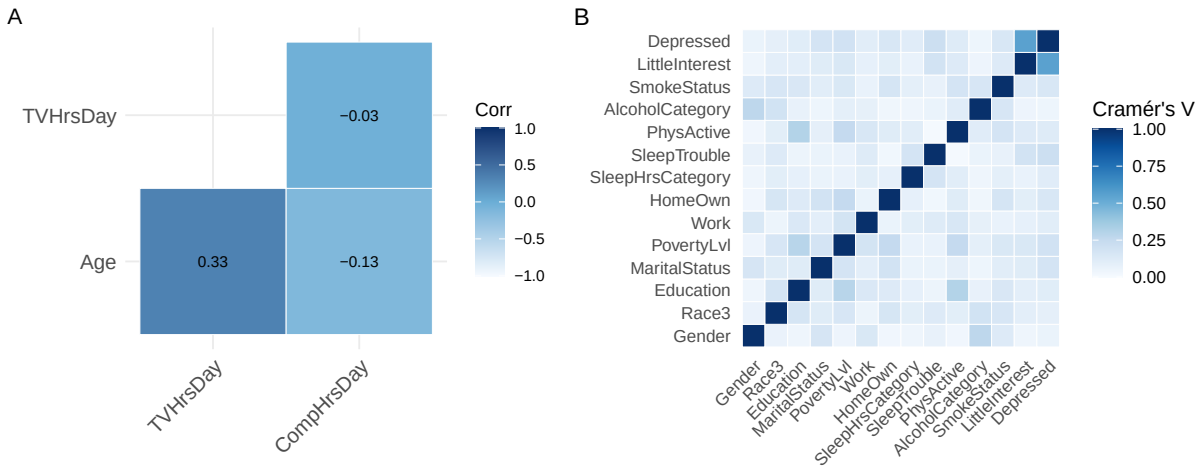


Figure 1: Correlation among Numeric Variables (Left) and Association among Categorical Variables (Right)

Table 3: Variance Inflation Factors (VIF)

Variable	GVIF	Df	GVIF^(1/(2*Df))
Gender	1.201	1	1.096
Age	1.910	1	1.382
Race3	1.479	5	1.040
Education	1.396	2	1.087
MaritalStatus	1.286	2	1.065
PovertyLvl	1.493	2	1.105
Work	1.630	2	1.130
HomeOwn	1.328	2	1.074
SleepHrsCategory	1.164	2	1.039
SleepTrouble	1.176	1	1.084
PhysActive	1.169	1	1.081
TVHrsDay	1.238	1	1.113
CompHrsDay	1.106	1	1.052
SmokeStatus	1.441	2	1.096
AlcoholCategory	1.290	2	1.066
LittleInterest	1.483	1	1.218
Depressed	1.580	1	1.257

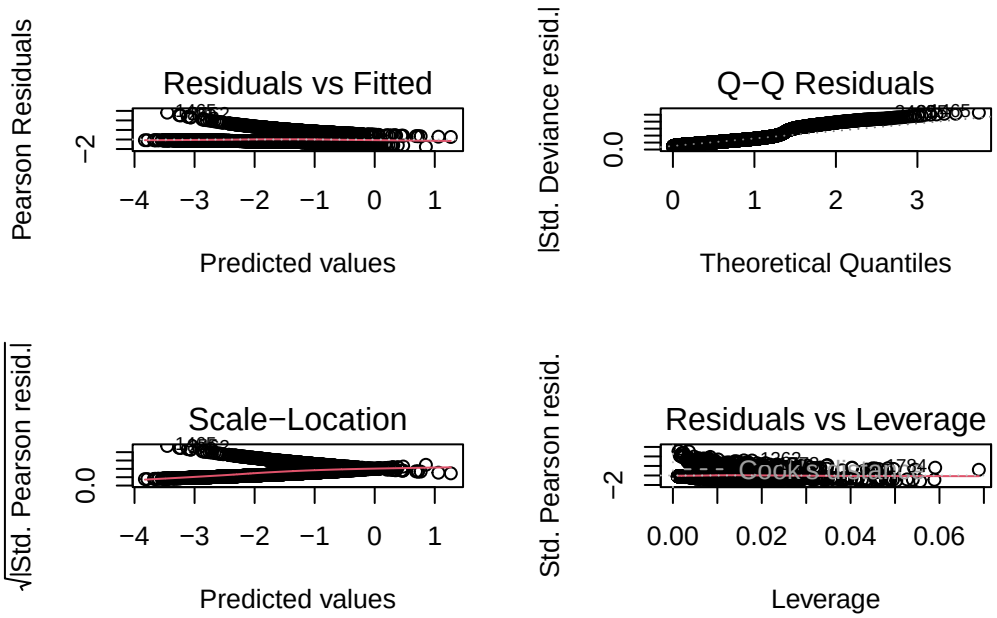


Figure 2: Diagnostic Plots

Table 4: Characteristics by Smoking Status

SmokeStatus	N	Mean Age	Median Age	Diabetes Prevalence (%)	Mean Age among Diabetics
Never	1244	59.4	58	16.6	63.9
Former	854	62.2	62	21.1	65.8
Current	775	53.0	51	15.0	57.8

Table 5: Characteristics by Alcohol Category

AlcoholCategory	No	Yes
Light-Moderate	1523	312
Non-drinker	720	169
Heavy	128	21

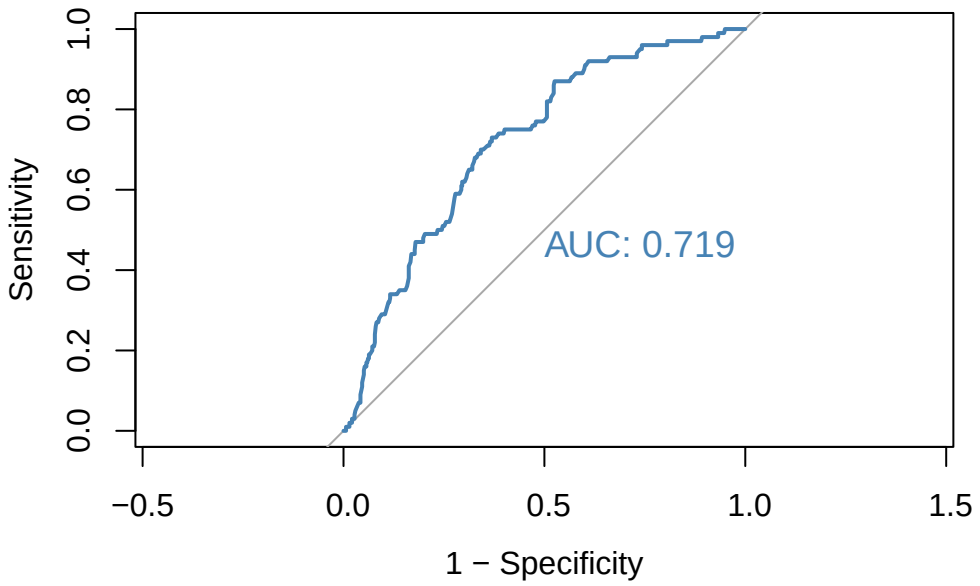


Figure 3: Receiver operating characteristic (ROC) curve for the balanced Random Forest classifier evaluated on the held-out test set.

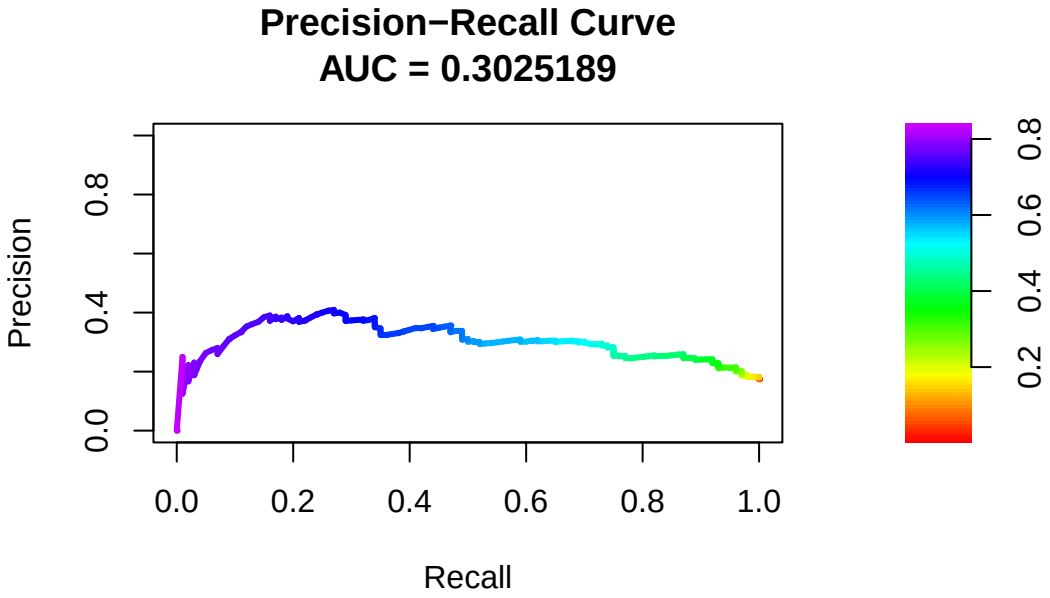


Figure 4: Precision–Recall curve for the Random Forest classifier evaluated on the test set.

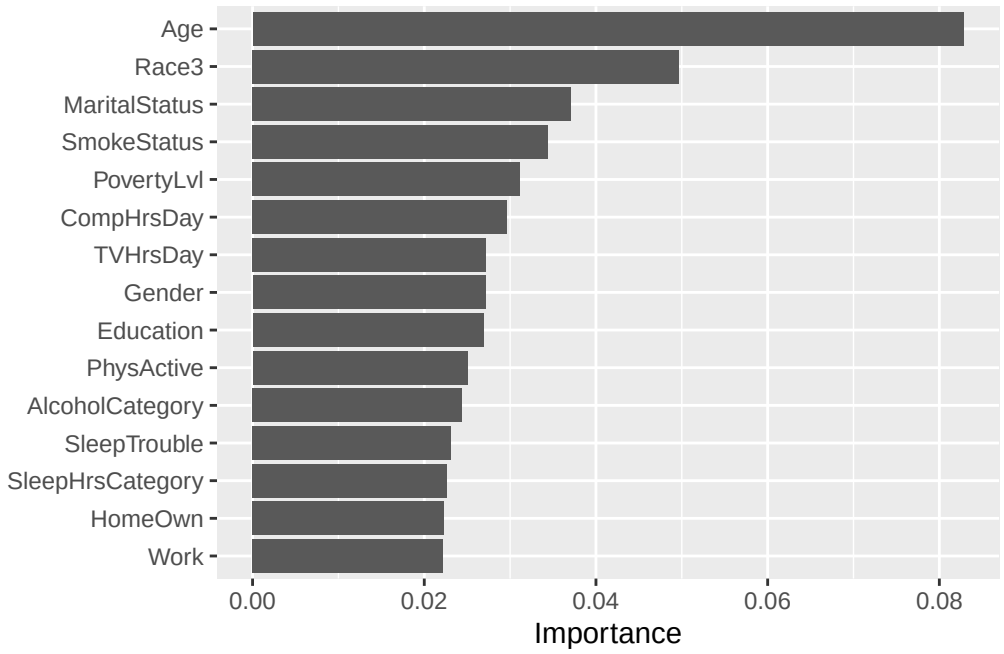


Figure 5: Permutation-based variable importance for the Random Forest classifier, ranked by decrease in ROC–AUC.